CSCI 7350 Data mining

Final Project2

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1. Which model performed best? Why?

Model	**Feature set**	**Accuracy**	**Macro F1**	**Weighted F1**
MLP	**Original Scaled**	**92.88%**	**93.94%**	**92.85%**
SVM	**Original Scaled**	**92.21%**	**93.43%**	**92.22%**
KNN	**Original Scaled**	**91.66%**	**92.93%**	**91.68%**

Among all evaluated models, the Multilayer Perceptron (MLP) trained on the original scaled feature set delivered the best overall classification performance, achieving the highest accuracy (92.88%), macro-averaged F1 score (93.94%), and weighted F1 score (92.85%). This superior performance suggests that MLP was particularly effective at capturing the complex nonlinear relationships among the Dry Bean dataset's morphological and geometric features, which likely enhanced its ability to distinguish between multiple bean classes. Unlike simpler models such as Decision Trees or AdaBoost, the neural network architecture was better suited to learning intricate class boundaries across the full feature space. Additionally, the use of the complete original feature set preserved all informative characteristics of the dataset, whereas feature selection methods such as variance thresholding, mutual information, and chi-square slightly reduced performance by removing potentially useful predictive information. SVM and KNN also performed strongly, but MLP consistently provided the most balanced and robust multiclass classification across both majority and minority classes. Overall, these results indicate that MLP on the original scaled data is the most effective model for this task, offering the best combination of predictive accuracy and generalization capability.

2. **Did your transformation (feature selection or PCA) improve performance?**

   Neither feature selection nor PCA improved overall classification performance compared to using the full original scaled feature set. Across all experiments, the original dataset consistently produced the highest accuracy, macro F1, and weighted F1 scores for nearly every model. The best-performing configuration was the MLP model on the original scaled data, achieving 92.88% accuracy and 92.85% weighted F1, while all feature selection methods—including variance thresholding, mutual information, and chi-square—resulted in slight performance declines. Variance thresholding generally preserved performance better than other selection approaches, but still underperformed relative to the full feature set. PCA, despite reducing the 16 original features to only 4 principal components while preserving 95% variance, caused a more noticeable drop in model effectiveness across all classifiers. This suggests that although PCA retained global variance, it discarded important class-specific discriminative information. Overall, the transformations reduced dimensionality and computational complexity but did not enhance predictive performance. Therefore, for this Dry Bean multiclass classification task, retaining the complete original feature set was the most effective strategy, indicating that the dataset's moderate dimensionality and informative features did not require aggressive dimensionality reduction.

3. **Which classes are most frequently confused?**

The confusion matrices indicate that the most frequently confused classes are primarily DERMASON and SIRA, which consistently show the highest rates of misclassification across nearly all models. For example, in the best-performing MLP model using the original scaled dataset, 52 SIRA samples were incorrectly classified as DERMASON, while 36 DERMASON samples were misclassified as SIRA. This suggests substantial overlap in their morphological characteristics, making them more difficult to distinguish. Similarly, BARBUNYA and CALI also exhibit notable confusion, with several BARBUNYA beans being predicted as CALI and vice versa, likely due to similarities in size and shape-related features. In contrast, the BOMBAY class was almost perfectly classified across all models, indicating that its feature distribution is highly distinct from the other bean types. Overall, the primary classification challenge lies in separating visually and morphologically similar medium-sized bean varieties, particularly DERMASON–SIRA and BARBUNYA–CALI, while more unique classes such as BOMBAY are easily identified. These confusion patterns highlight areas where advanced feature engineering or class-specific optimization may further improve model performance.

4. What differences do you observe between macro and weighted F1 scores?

The differences between macro-averaged F1 and weighted F1 scores reveal important insights about class imbalance and model consistency across the Dry Bean dataset. In nearly all models, the weighted F1 score is slightly higher than or very close to the macro F1 score, reflecting the influence of larger classes such as DERMASON and SIRA, which contribute more heavily to the weighted metric due to their greater sample sizes. Weighted F1 emphasizes performance on the majority classes, while macro F1 treats all classes equally, regardless of class frequency. The relatively small gap between these two metrics for top-performing models such as MLP and SVM suggests that these models performed consistently well across both majority and minority classes, including smaller categories like BOMBAY and BARBUNYA. However, larger discrepancies in weaker models, such as AdaBoost, indicate uneven performance, where the model performs better on dominant classes but struggles more with minority classes. Therefore, macro F1 serves as a better indicator of balanced multiclass performance, while weighted F1 reflects overall practical performance considering class distribution. In this study, the close alignment of macro and weighted F1 scores for the best models demonstrates strong generalization and fairness across all bean categories.